CSE 311:
Data Communication

Instructor:

Dr. Md. Monirul Islam

Signals and systems

Topics

- Definition, classification and properties of signals
- Examples of some useful signals
- Fourier Series of Periodic Signals

Alternate representation of $u(t)$ in terms of $\delta(t)$

$$\int_{-\infty}^{t} \delta(\tau)\, d\tau$$

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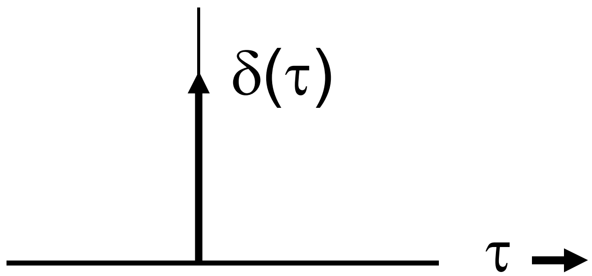
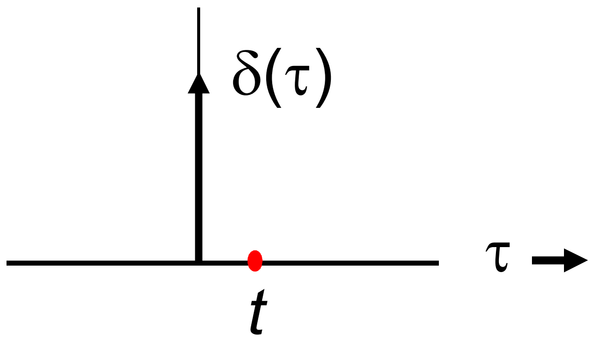


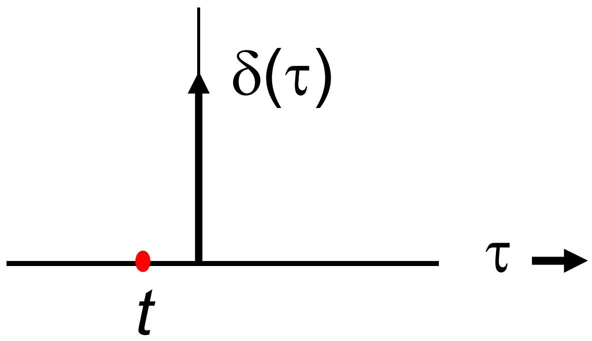
Alternate representation of $u(t)$ in terms of $\delta(t)$

$$\int_{-\infty}^{t} \delta(\tau)\, d\tau = 1$$



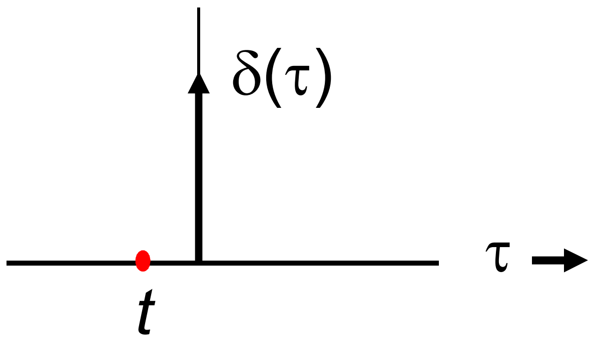
Alternate representation of $u(t)$ in terms of $\delta(t)$

$$\int_{-\infty}^{t} \delta(\tau)\, d\tau = 0$$



Alternate representation of $u(t)$ in terms of $\delta(t)$

$$\int_{-\infty}^{t} \delta(\tau)\, d\tau = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$



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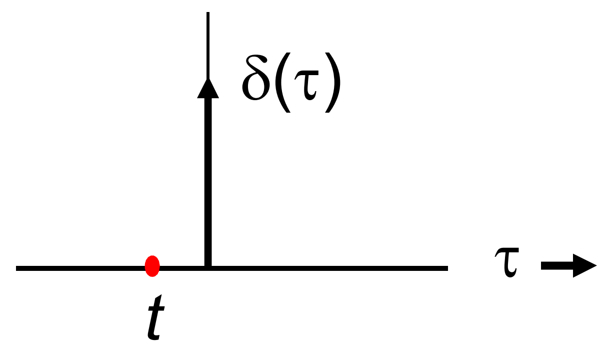
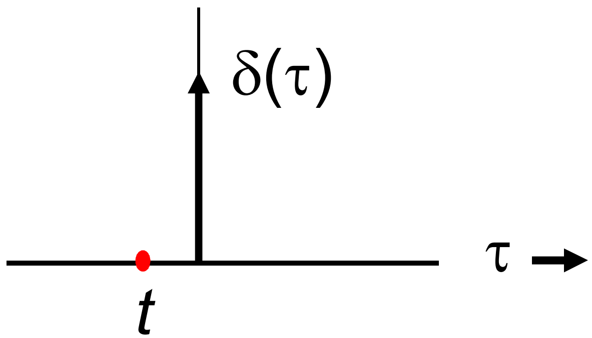


$$u(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases}$$

Alternate representation of $u(t)$ in terms of $\delta(t)$

$$\int_{-\infty}^{t} \delta(\tau)\, d\tau = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$
$$= u(t)$$



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Alternate representation of $u(t)$ in terms of $\delta(t)$

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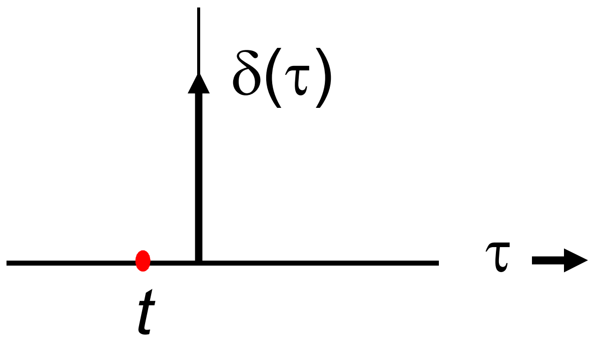

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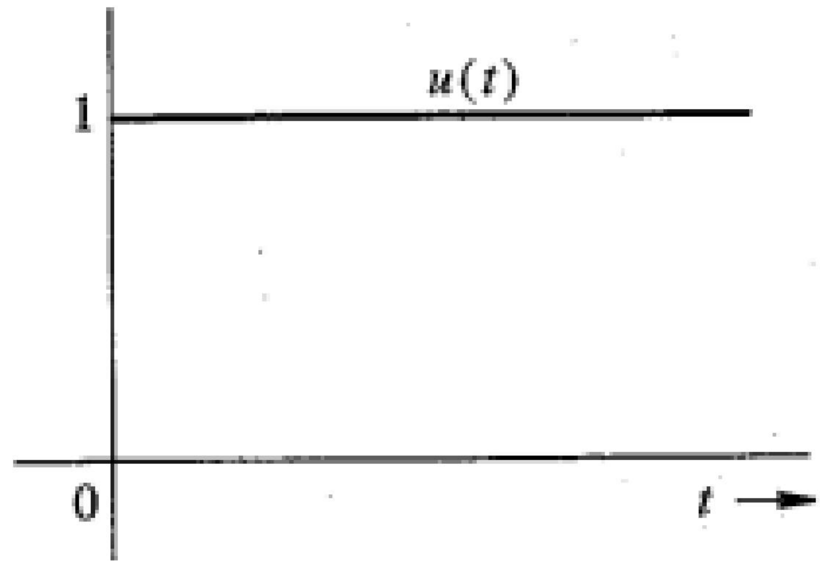
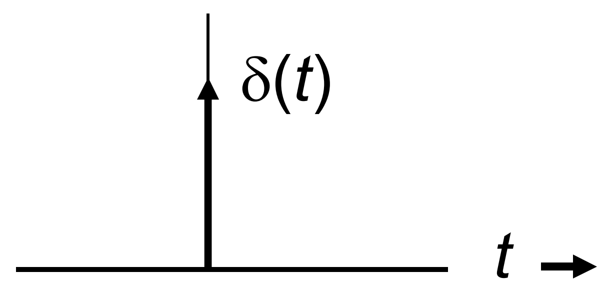


$$\frac{du}{dt} = \delta(t)$$



Fourier Series of a periodic signal

‘Any function that *periodically repeats itself* can be expressed as the *sum of sines and/or cosines* of different frequencies’

- Joseph Fourier

Fourier Series of a periodic signal

'Any function that *periodically repeats itself* can be expressed as the *sum of sines and/or cosines* of different frequencies'

- Joseph Fourier

$$g(t) = \sum_{n=-\infty}^{n=\infty} D_n e^{jn\omega_0 t}$$

$$= \sum_{n=-\infty}^{n=\infty} D_n e^{jn2\pi f_0 t}$$

with, T_0 = period of $g(t)$
f_0 = frequency of $g(t)$

where, $D_n = \dfrac{1}{T_0} \displaystyle\int_{T_0} g(t) e^{-jn2\pi f_0 t}$

Fourier Series

Fourier says:

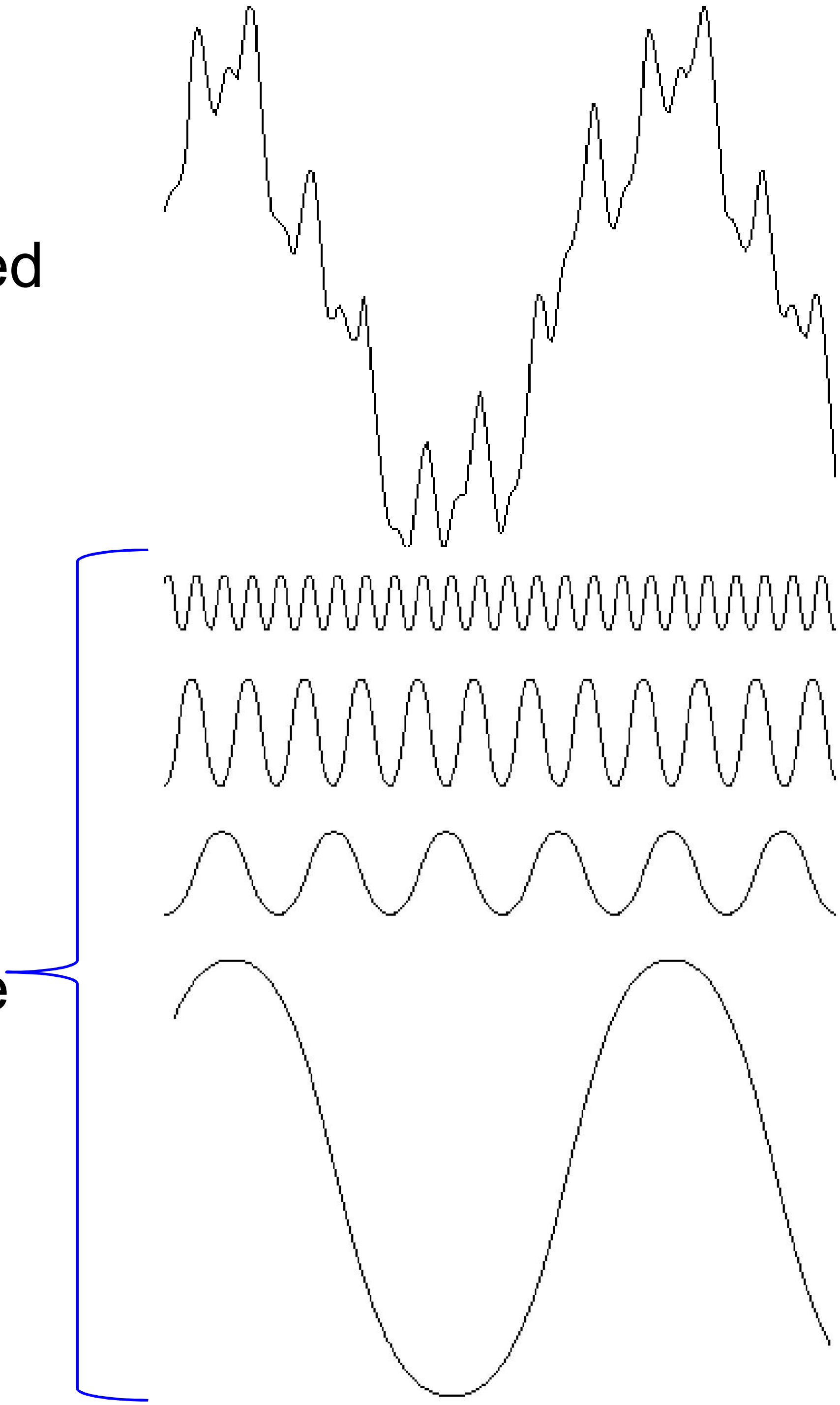
Any function that *periodically repeats itself* can be expressed as the *sum of sines and/or cosines* of different frequencies

- Doesn’t matter how complicated the function is!
- The summation is called *Fourier Series*

Fourier Series

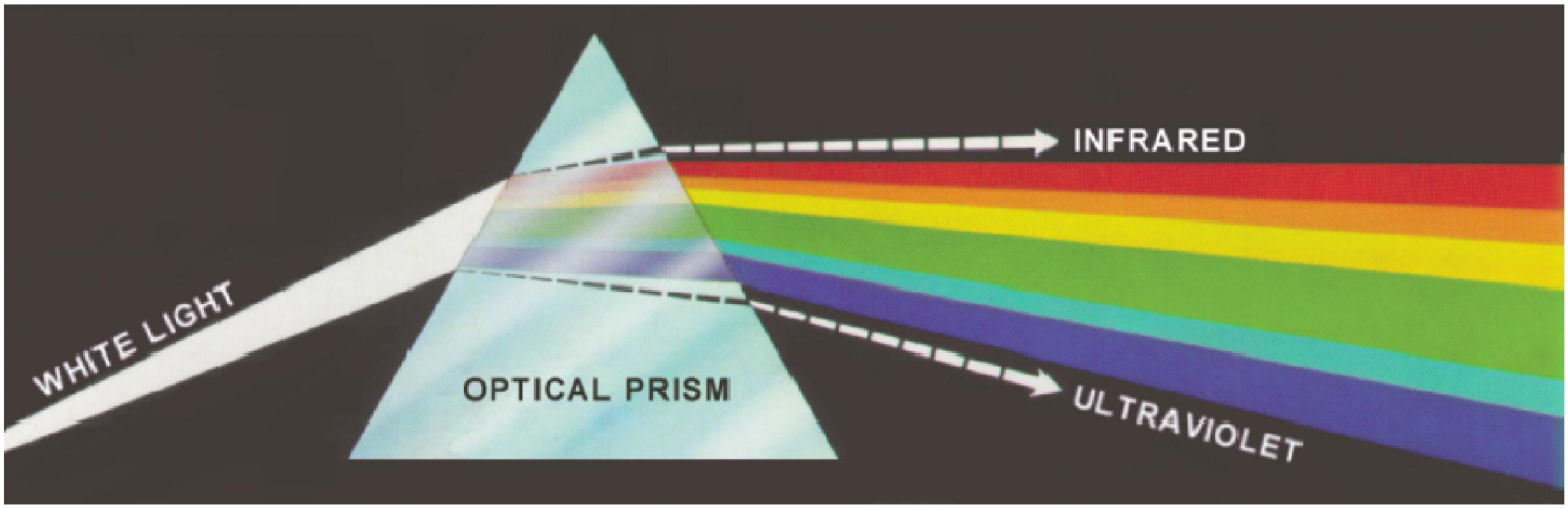
A complicated but periodic function

Complicated periodic function



Simpler sine/cosine functions

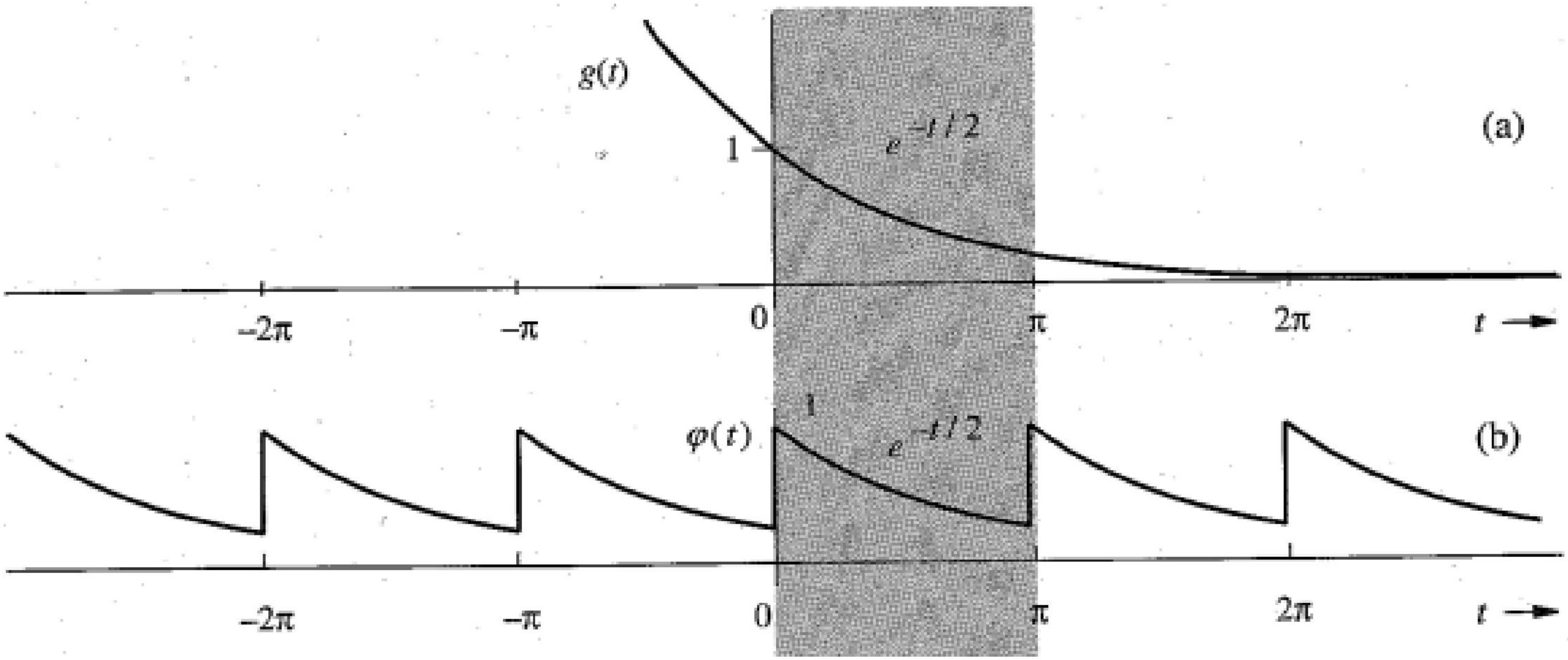
Fourier Series and Light Spectrum



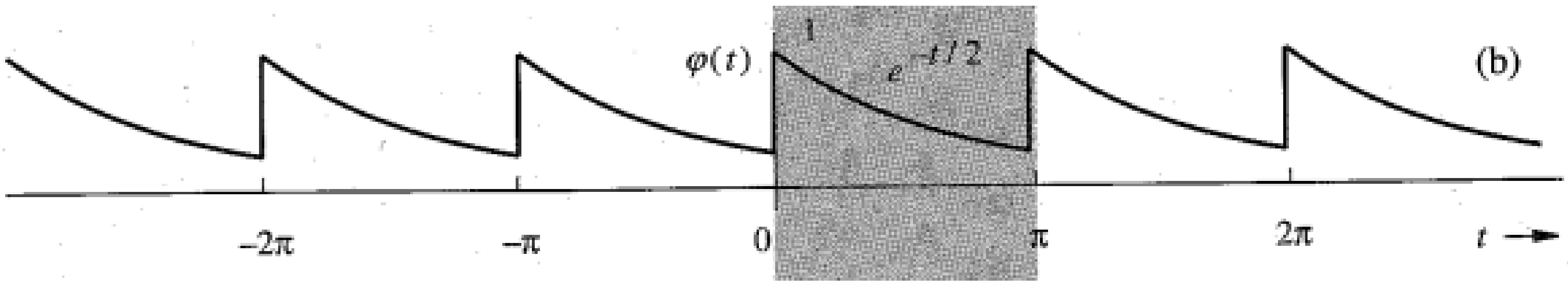
A complicated function/signal

Simpler functions/signals

Representing a Periodic signal using Fourier Series



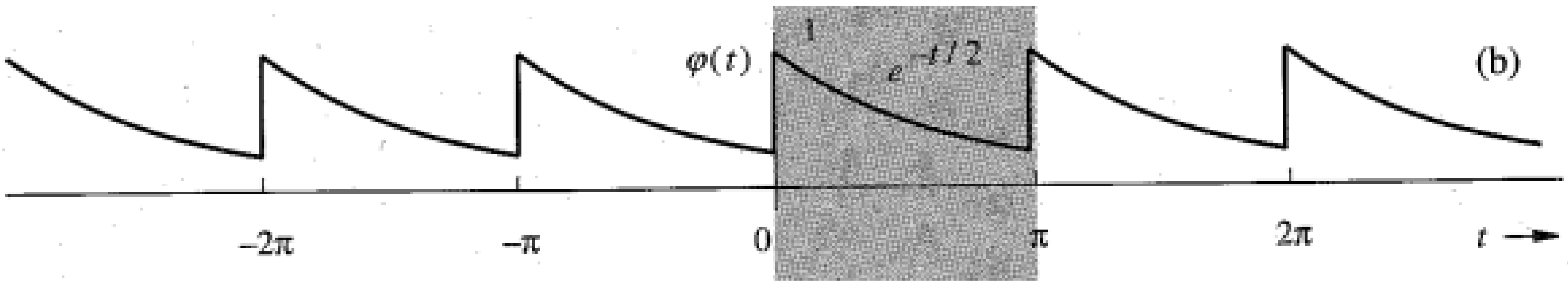
Representing a Periodic signal using Fourier Series



$$\phi(t) = \sum_{n=-\infty}^{n=\infty} D_n e^{jn\omega_0 t} = \sum_{n=-\infty}^{n=\infty} D_n e^{j2nt}$$

Here, $T_0 = \pi$ and $\omega_0 = \frac{2\pi}{T_0} = 2 \text{ rad/s}$

Representing a Periodic signal using Fourier Series

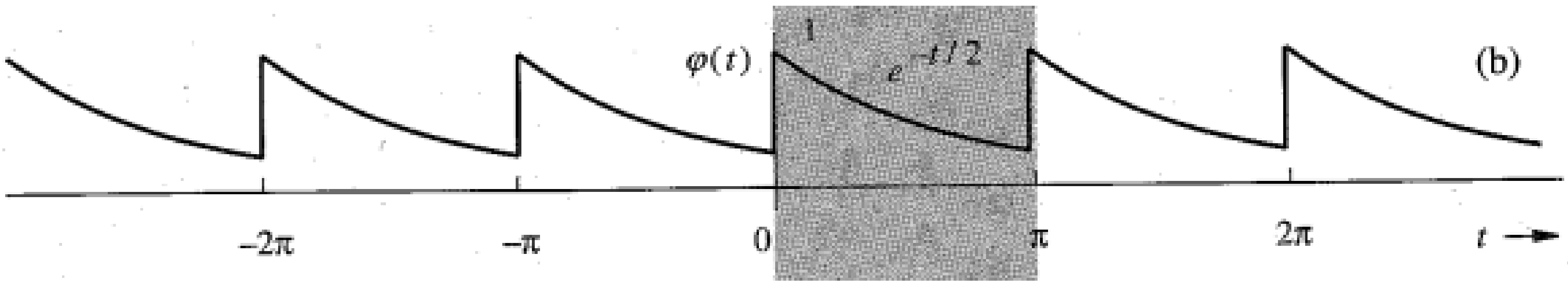


$$D_n = \frac{1}{T_0}\int_{T_0} \varphi(t) e^{-j2nt}\, dt = \frac{1}{\pi}\int_0^{\pi} e^{-t/2}\, e^{-j2nt}\, dt$$

$$= \frac{1}{\pi}\int_0^{\pi} e^{-(\frac{1}{2}+j2n)t}\, dt = \frac{-1}{\pi\left(\frac{1}{2}+j2n\right)} e^{-(\frac{1}{2}+j2n)t}\Big|_0^{\pi}$$

$$= \frac{0.504}{1+j4n}$$

Representing a Periodic signal using Fourier Series



$$D_n = \frac{0.504}{1 + j4n}$$

$$\phi(t) = \sum_{n=-\infty}^{n=\infty} D_n e^{j2nt}$$

$$\varphi(t) = 0.504 \sum_{n=-\infty}^{\infty} \frac{1}{1 + j4n} e^{j2nt}$$

$$= 0.504 \left[1 + \frac{1}{1+j4} e^{j2t} + \frac{1}{1+j8} e^{j4t} + \frac{1}{1+j12} e^{j6t} + \cdots \right.$$

$$\left. + \frac{1}{1-j4} e^{-j2t} + \frac{1}{1-j8} e^{-j4t} + \frac{1}{1-j12} e^{-j6t} + \cdots \right]$$

Representing a Periodic signal using Fourier Series

- D_n are exponential Fourier spectra
- D_n are complex though $\phi(t)$ is a real periodic
- D_n and D_{-n} are complex conjugates

$$D_n = \frac{0.504}{1 + j4n}$$

$$\begin{aligned}\varphi(t) &= 0.504 \sum_{n=-\infty}^{\infty} \frac{1}{1 + j4n} e^{j2nt} \\ &= 0.504 \left[1 + \frac{1}{1 + j4} e^{j2t} + \frac{1}{1 + j8} e^{j4t} + \frac{1}{1 + j12} e^{j6t} + \cdots \right. \\ &\qquad \left. + \frac{1}{1 - j4} e^{-j2t} + \frac{1}{1 - j8} e^{-j4t} + \frac{1}{1 - j12} e^{-j6t} + \cdots \right]\end{aligned}$$

Representing a Periodic signal using Fourier Series

- D_n are complex though $\phi(t)$ is a real periodic
  - $D_n = |D_n| e^{j\angle D_n}$

Representing a Periodic signal using Fourier Series

- D_n are complex though $\phi(t)$ is a real periodic
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- D_n and D_{-n} are complex conjugates
  - $|D_n| = |D_{-n}|$
  - $\angle D_n = \theta_n$ and $\angle D_{-n} = -\theta_n$

Representing a Periodic signal using Fourier Series

- D_n are complex though $\phi(t)$ is a real periodic
  - $D_n = |D_n| e^{j\angle D_n}$

- D_n and D_{-n} are complex conjugates
  - $|D_n| = |D_{-n}|$
  - $\angle D_n = \theta_n \quad \text{and} \quad \angle D_{-n} = -\theta_n$
- Therefore, $D_n = |D_n| e^{j\theta_n} \quad \text{and} \quad D_{-n} = |D_n| e^{-j\theta_n}$

Representing a Periodic signal using Fourier Series

$$|D_n| = |D_{-n}|$$

$$\angle D_n = \theta_n \qquad \text{and} \qquad \angle D_{-n} = -\theta_n$$

- $|D_n|$ are amplitudes and $\angle D_n$ are angles of exponential spectra
- $|D_n|$ vs. f is an even function
- $\angle D_n$ vs. f is an odd function

Representing a Periodic signal using Fourier Series

$$D_n = \frac{0.504}{1 + j4n}$$

$$D_0 = 0.504$$

$$D_1 = \frac{0.504}{1 + j4} = 0.122e^{-j75.96^\circ} \Longrightarrow |D_1| = 0.122 \qquad \angle D_1 = -75.96^\circ$$

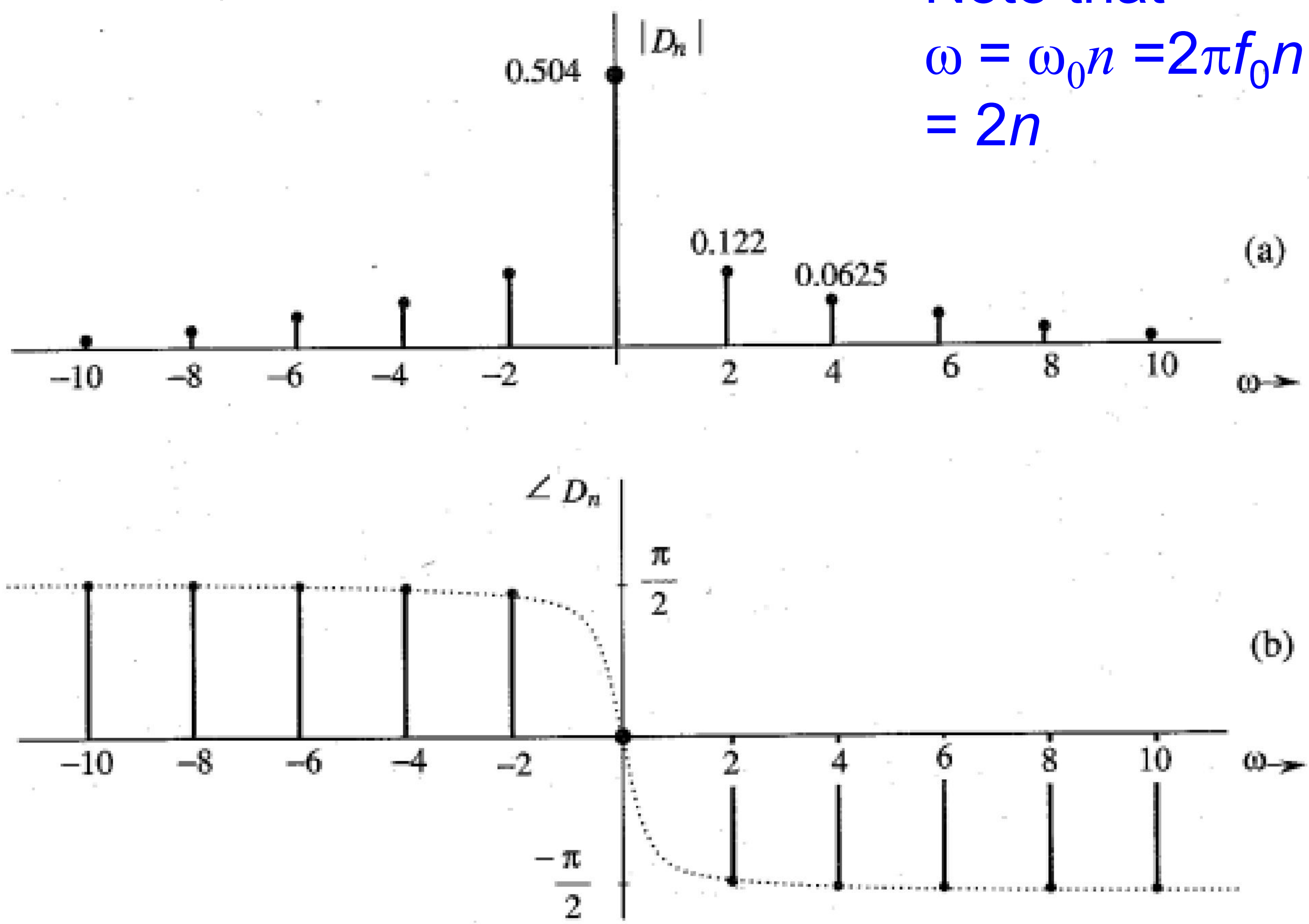
$$D_{-1} = \frac{0.504}{1 - j4} = 0.122e^{j75.96^\circ} \Longrightarrow |D_{-1}| = 0.122 \qquad \angle D_{-1} = 75.96^\circ$$

$$D_2 = \frac{0.504}{1 + j8} = 0.0625e^{-j82.87^\circ} \Longrightarrow |D_2| = 0.0625 \qquad \angle D_2 = -82.87^\circ$$

$$D_{-2} = \frac{0.504}{1 - j8} = 0.0625e^{j82.87^\circ} \Longrightarrow |D_{-2}| = 0.0625 \qquad \angle D_{-2} = 82.87^\circ$$

Representing a Periodic signal using Fourier Series

Note that
$\omega = \omega_0 n = 2\pi f_0 n$
$= 2n$



Representing a Periodic signal using Fourier Series

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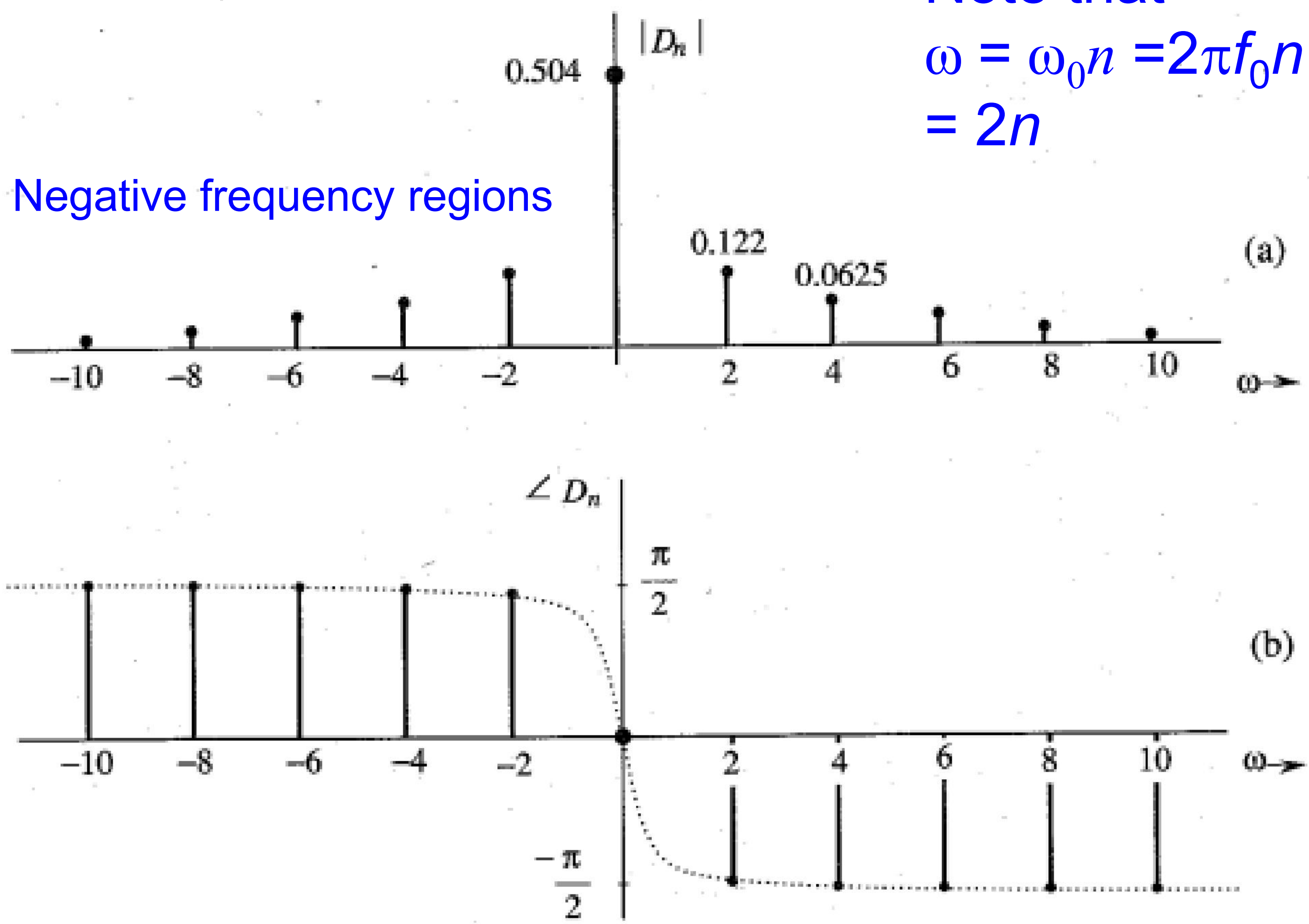
Negative frequency regions



Meaning of negative frequencies

Consider this real periodic function with frequency f_0

$$\cos(-\omega_0 t + \theta) = \cos(\omega_0 t - \theta)$$

Note that angular frequency is $\omega_0 = 2\pi f_0$

Meaning of negative frequencies

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Here, angular frequency is $|\omega_0|$, a positive quantity which only indicates rate of sinusoidal variation

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$$\cos(-\omega_0 t + \theta) = \cos(\omega_0 t - \theta)$$

Here, angular frequency is $|\omega_0|$, a positive quantity which only indicates rate of sinusoidal variation

Cannot explain negative frequency

Meaning of negative frequencies

Consider this complex sinusoid function with frequency f_0

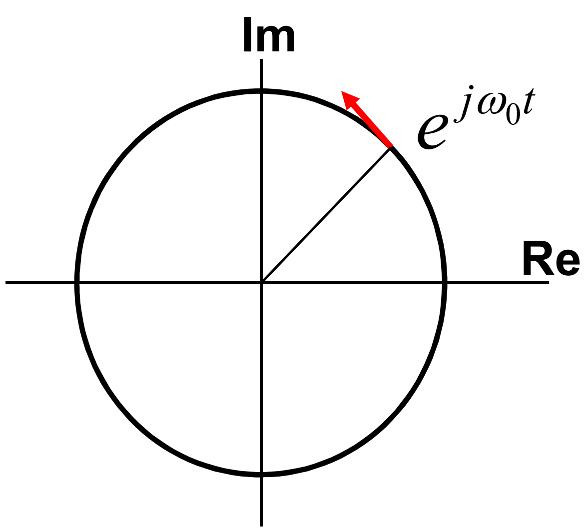
$$e^{\pm j\omega_0 t} = \cos\ \omega_0 t \pm j \sin\ \omega_0 t$$

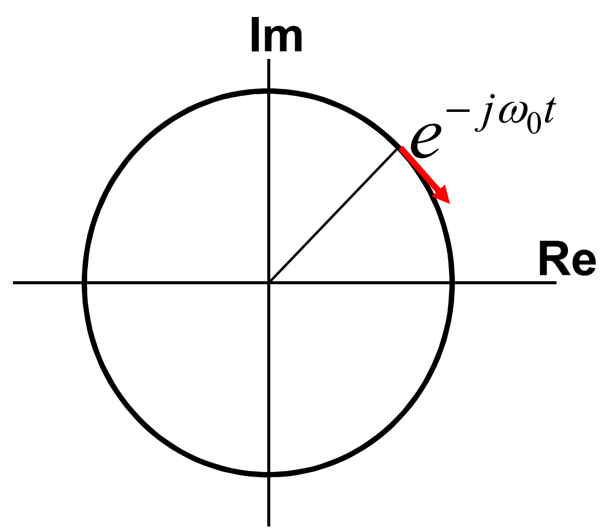
It both describes the rate variation, $|\omega_0|$, and direction ($\pm$)

Meaning of negative frequencies

Consider this complex sinusoid function with frequency f_0

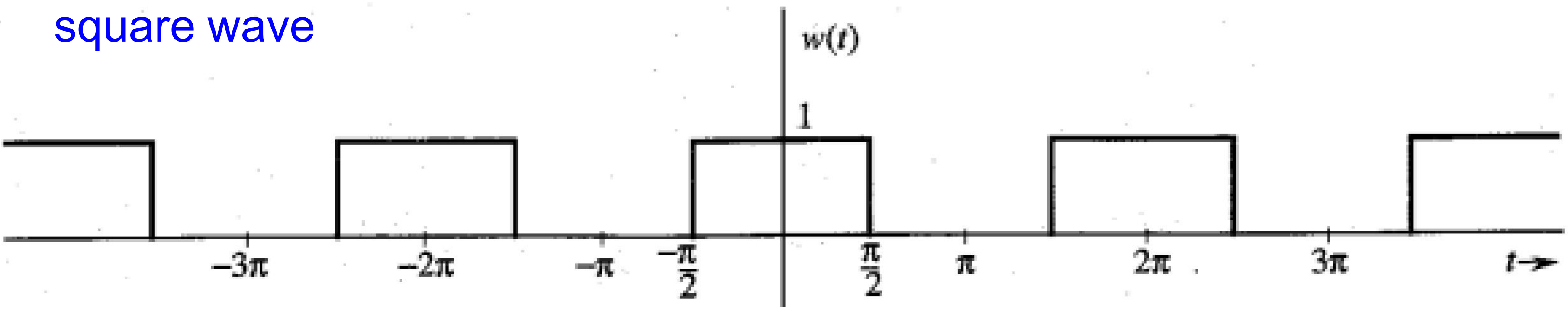
$$e^{\pm j\omega_0 t} = \cos\ \omega_0 t \pm j \sin\ \omega_0 t$$





Fourier series of Some Useful Signals

Periodic
square wave



Period, $T_0 = 2\pi$
Frequency, $f_0 = 1/T_0 = 1/\ 2\pi$

Fourier series of Some Useful Signals

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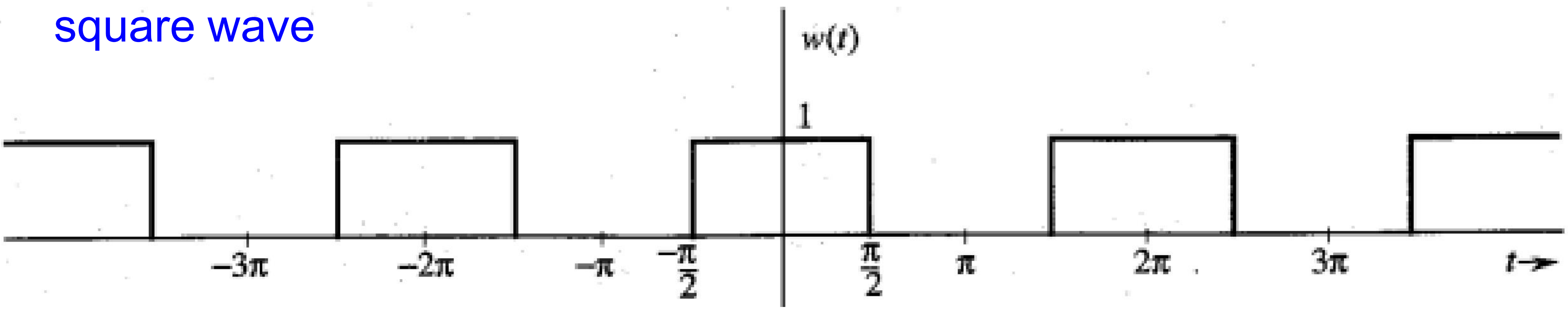
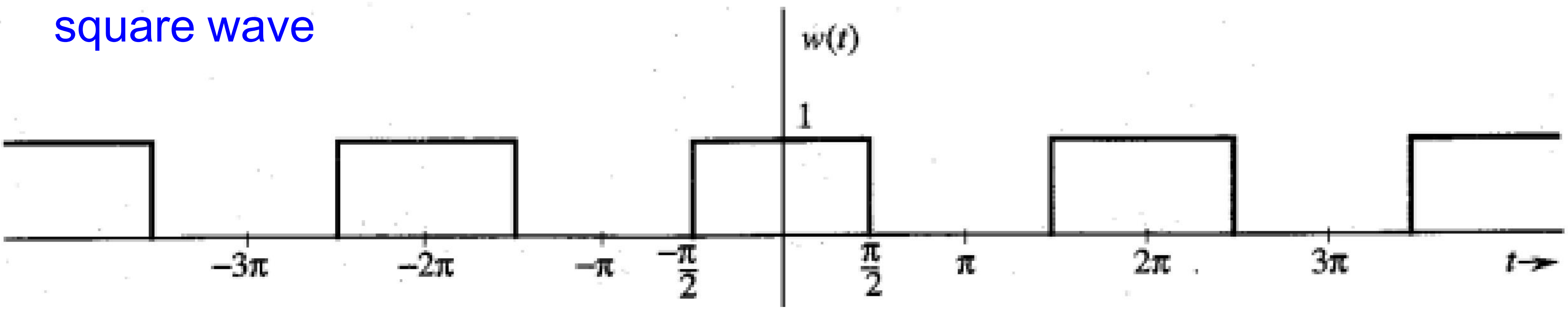


Period, $T_0 = 2\pi$
Frequency, $f_0 = 1/T_0 = 1/\ 2\pi$

$$w(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}$$ where, $\omega_0 = 2\pi f_0$

Fourier series of Some Useful Signals

Periodic
square wave

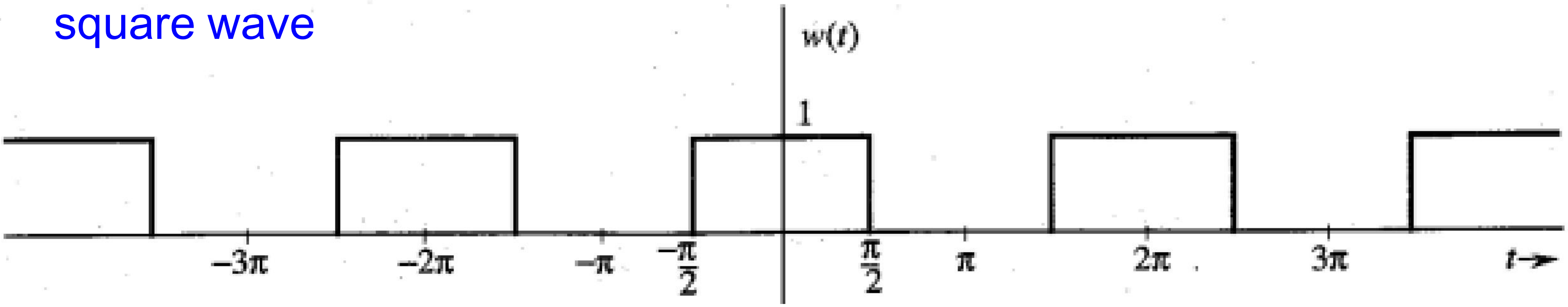


Period, $T_0 = 2\pi$
Frequency, $f_0 = 1/T_0 = 1/2\pi$

$$D_n = \frac{1}{T_0} \int_{T_0} w(t) e^{-jn\omega_0 t}\, dt$$

Fourier series of Some Useful Signals

Periodic
square wave

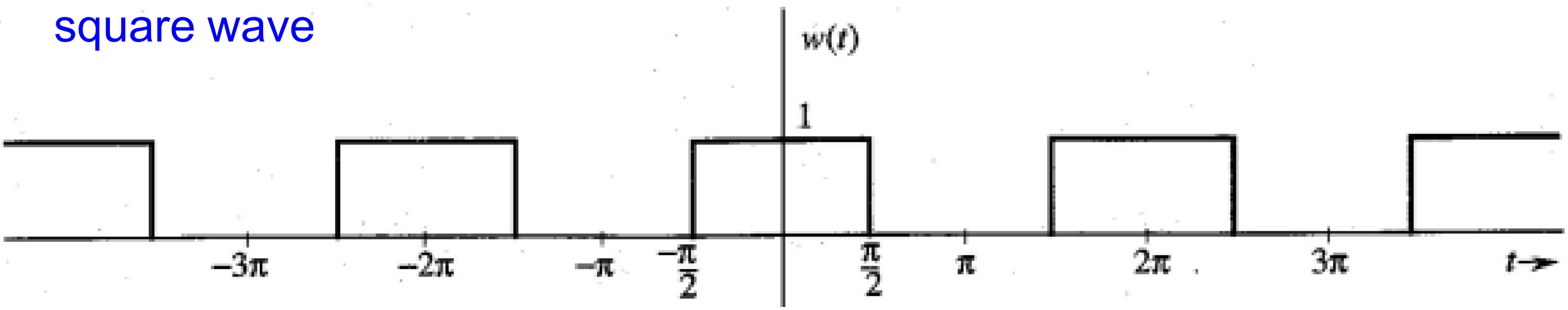


Period, $T_0 = 2\pi$
Frequency, $f_0 = 1/T_0 = 1/2\pi$

$$D_n = \frac{1}{T_0}\int_{T_0} w(t)e^{-jn\omega_0 t}\,dt = \frac{1}{T_0}\int_{-T_0/4}^{T_0/4} e^{-jn\omega_0 t}\,dt$$

Fourier series of Some Useful Signals

Periodic
square wave

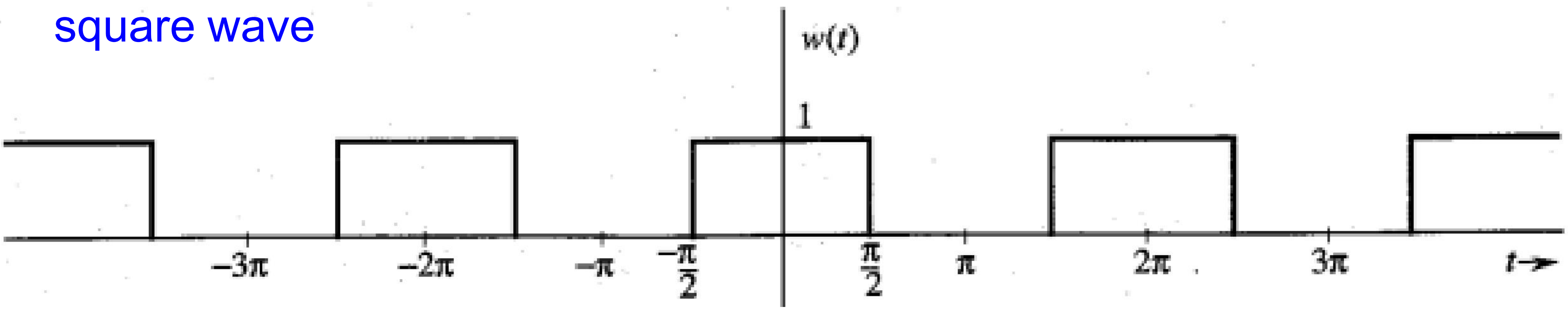


Period, $T_0 = 2\pi$
Frequency, $f_0 = 1/T_0 = 1/2\pi$

$$D_n = \frac{1}{T_0}\int_{T_0} w(t)e^{-jn\omega_0 t}\,dt = \frac{1}{T_0}\int_{-T_0/4}^{T_0/4} e^{-jn\omega_0 t}\,dt$$
$$= \frac{1}{-jn\omega_0 T_0}\left(e^{-jn\omega_0 T_0/4} - e^{jn\omega_0 T_0/4}\right)$$

Fourier series of Some Useful Signals

Periodic
square wave



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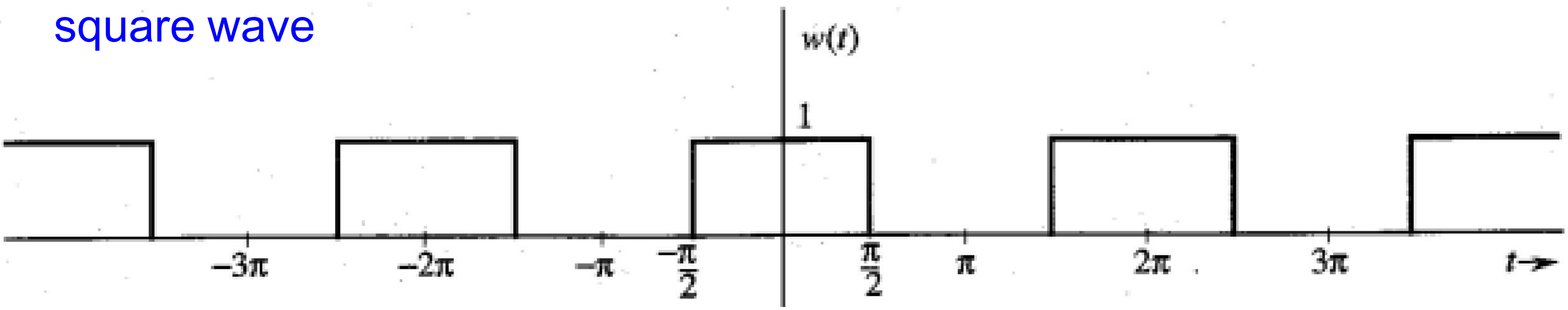
$$D_n = \frac{1}{T_0}\int_{T_0} w(t)e^{-jn\omega_0 t}\,dt = \frac{1}{T_0}\int_{-T_0/4}^{T_0/4} e^{-jn\omega_0 t}\,dt$$

$$= \frac{1}{-jn\omega_0 T_0}\left(e^{-jn\omega_0 T_0/4} - e^{jn\omega_0 T_0/4}\right)$$

$$= \frac{2}{n\omega_0 T_0}\sin\left(\frac{n\omega_0 T_0}{4}\right) = \frac{1}{n\pi}\sin\left(\frac{n\pi}{2}\right)$$

Fourier series of Some Useful Signals

Periodic
square wave



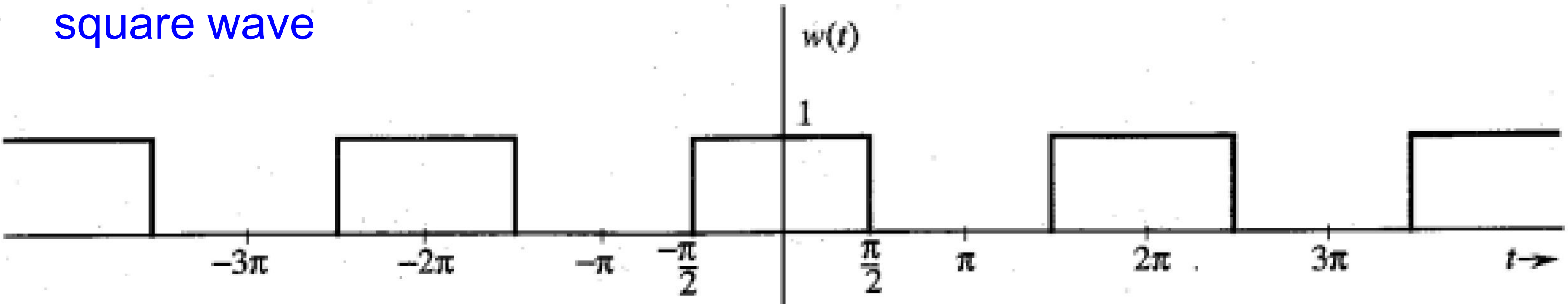
Period, $T_0 = 2\pi$
Frequency, $f_0 = 1/T_0 = 1/2\pi$

$$
\begin{aligned}
D_n &= \frac{1}{T_0}\int_{T_0} w(t)e^{-jn\omega_0 t}\,dt = \frac{1}{T_0}\int_{-T_0/4}^{T_0/4} e^{-jn\omega_0 t}\,dt \\
&= \frac{1}{-jn\omega_0 T_0}\left(e^{-jn\omega_0 T_0/4} - e^{jn\omega_0 T_0/4}\right) \\
&= \frac{2}{n\omega_0 T_0}\sin\left(\frac{n\omega_0 T_0}{4}\right) = \frac{1}{n\pi}\sin\left(\frac{n\pi}{2}\right)
\end{aligned}
$$

D_0 is undefined

Fourier series of Some Useful Signals

Periodic
square wave



Period, $T_0 = 2\pi$
Frequency, $f_0 = 1/T_0 = 1/2\pi$

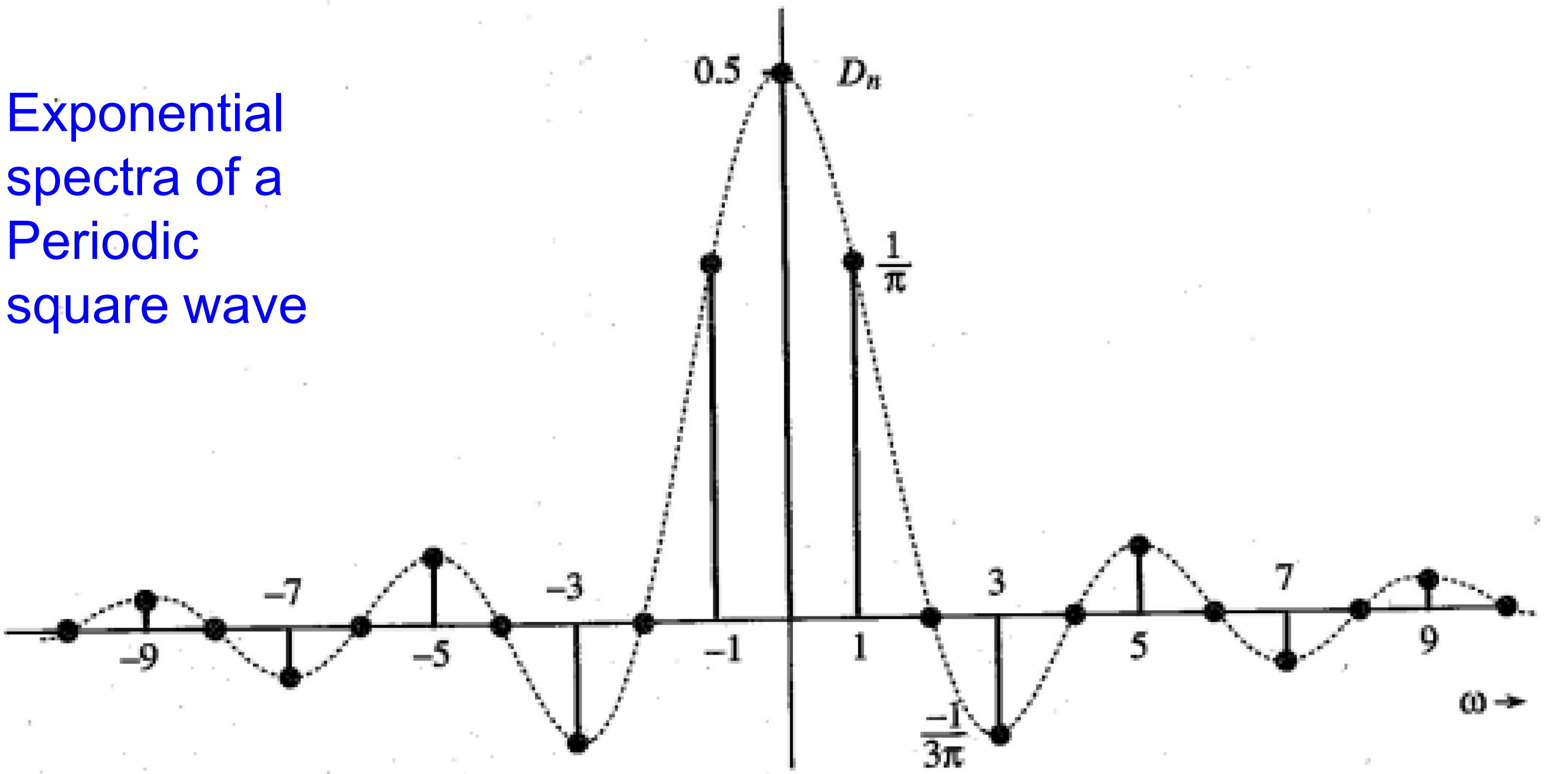
$$D_n = \frac{1}{T_0}\int_{T_0} w(t)e^{-jn\omega_0 t}\,dt = \frac{1}{n\pi}\sin\left(\frac{n\pi}{2}\right)$$

$$D_0 = \frac{1}{T_0}\int_{T_0} w(t)dt = \frac{1}{2}$$

Fourier series of Some Useful Signals

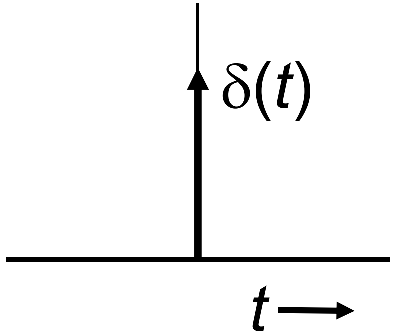
$$D_n = \frac{1}{n\pi}\sin\left(\frac{n\pi}{2}\right) \qquad D_0 = \frac{1}{2}$$

Exponential spectra of a Periodic square wave



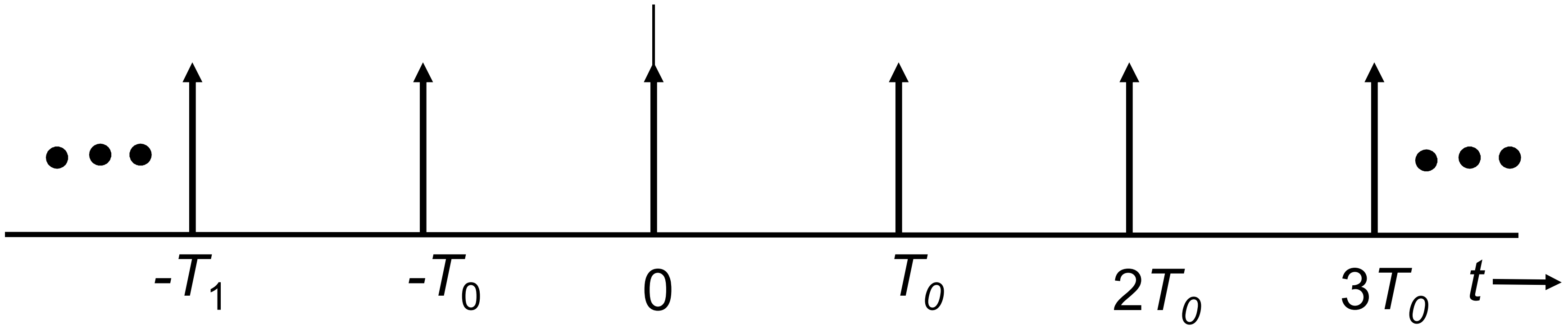
Fourier Series of Unit Impulse Train

Impulse signal



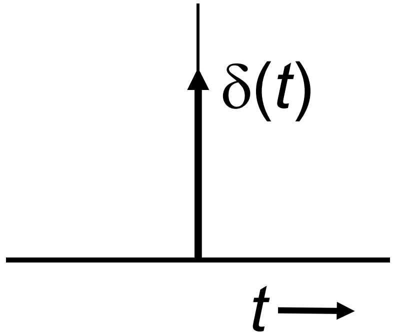
Impulse train

$$g(t) = \delta_{T_0}(t)$$



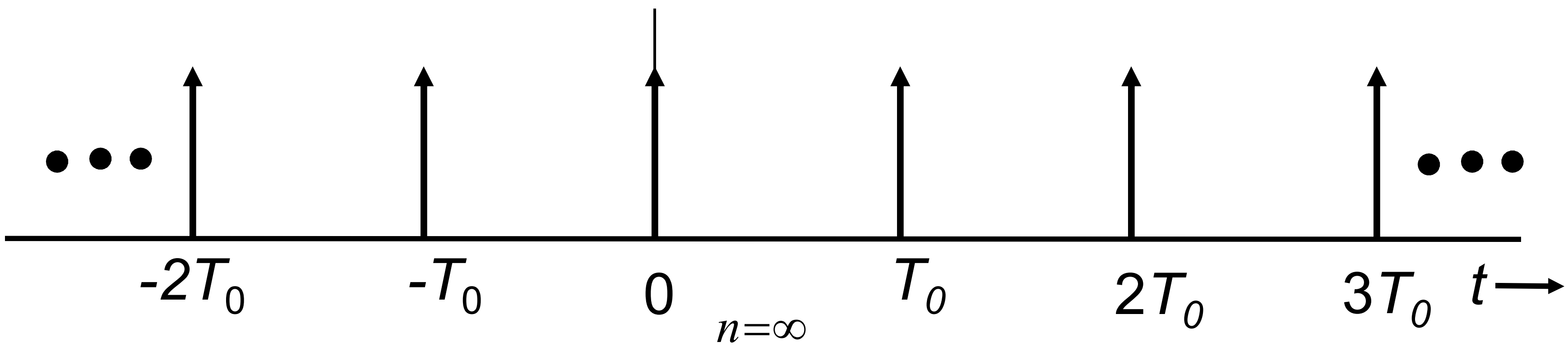
Fourier Series of Unit Impulse Train

Impulse signal



Impulse train

$$g(t) = \delta_{T_0}(t)$$

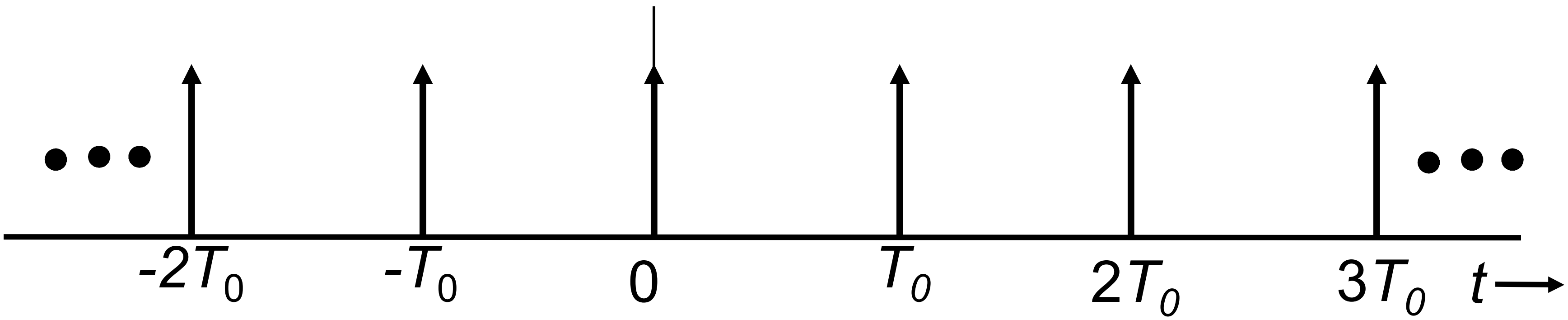


Equation of an Impulse train

$$s_{T_0}(t) = \sum_{n=-\infty}^{n=\infty} \delta(t - nT_0)$$

Fourier Series of Unit Impulse Train

$$g(t) = \delta_{T_0}(t)$$

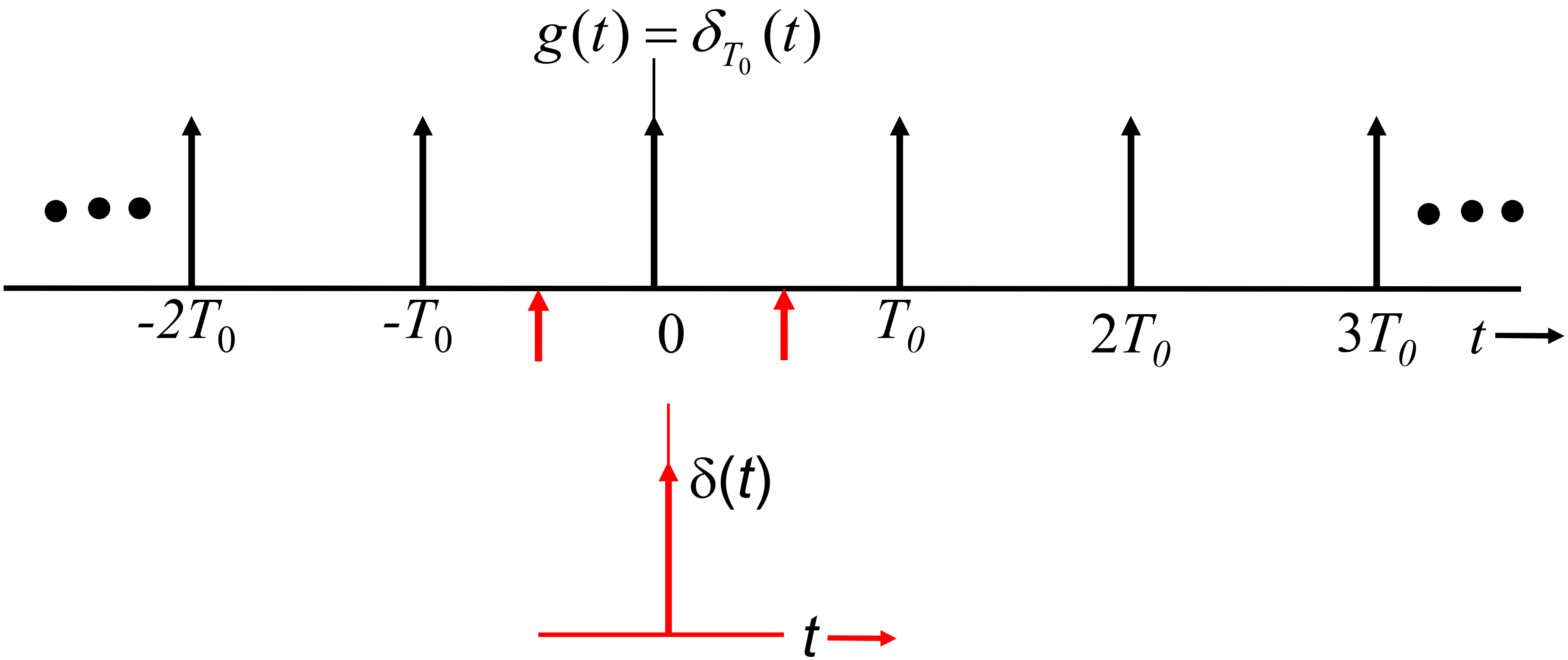


As it is a periodic signal with period T_0

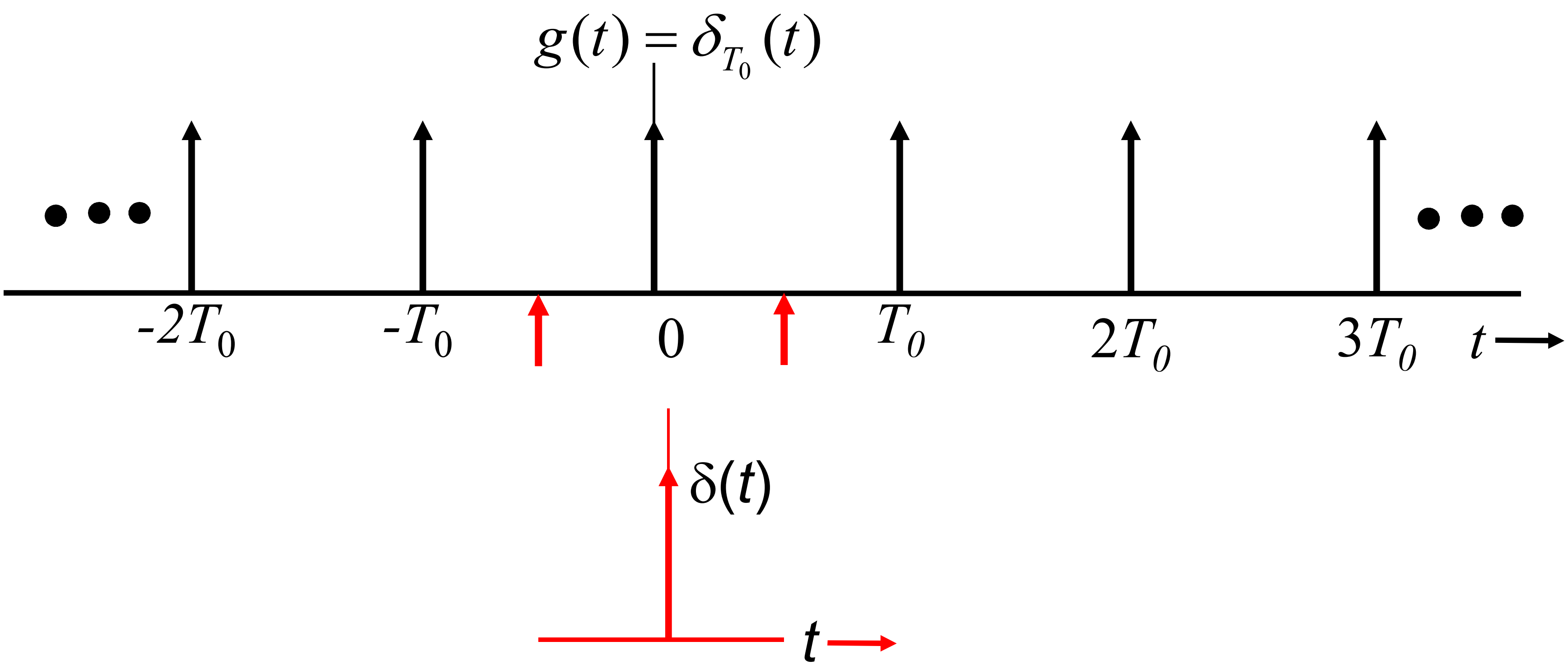
$$\delta_{T_0}(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t} \qquad \omega_0 = \frac{2\pi}{T_0}$$

where, $D_n = \dfrac{1}{T_0} \displaystyle\int_{T_0} \delta_{T_0}(t) e^{-jn\omega_0 t}\, dt$

Fourier Series of Unit Impulse Train

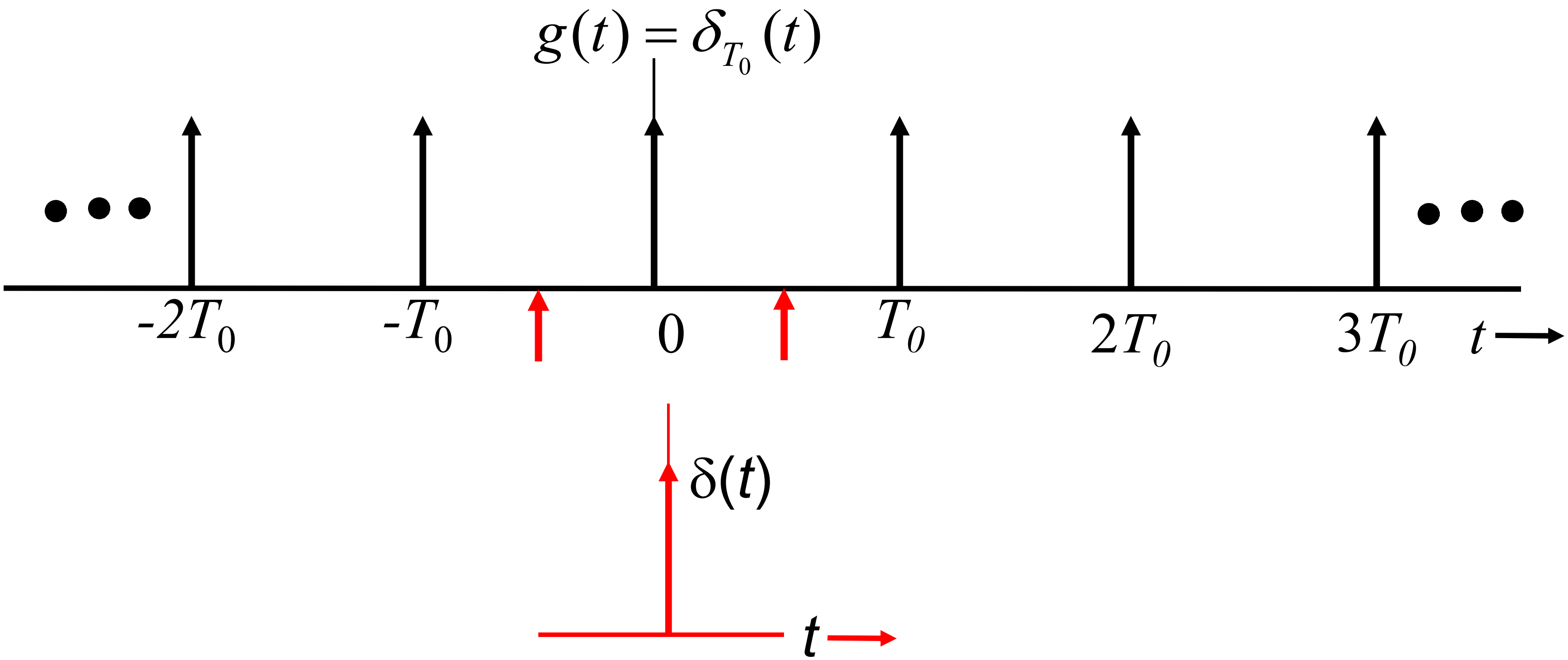


Fourier Series of Unit Impulse Train



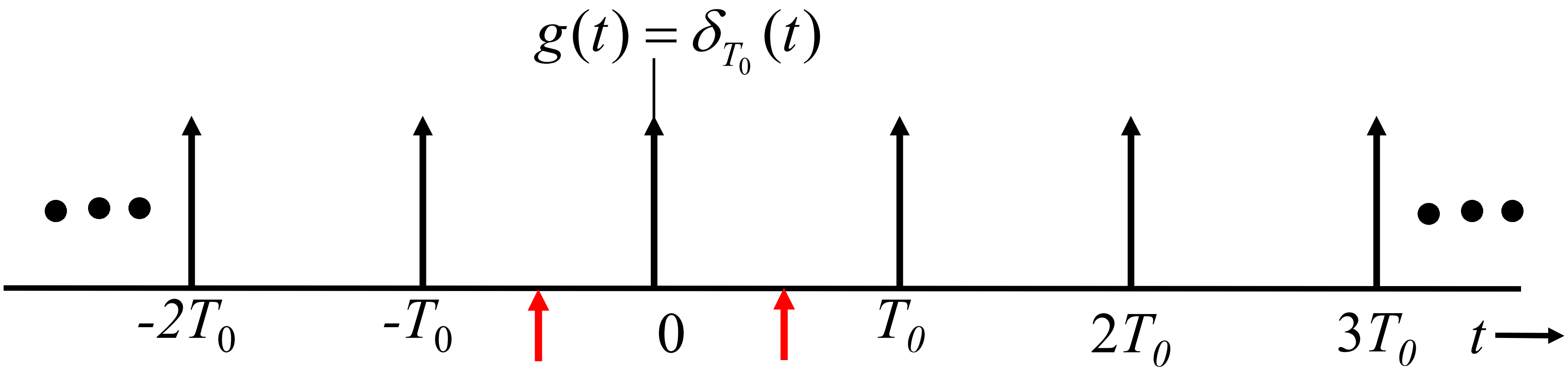
$$D_n = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} \delta(t) e^{-jn\omega_0 t}\, dt$$

Fourier Series of Unit Impulse Train

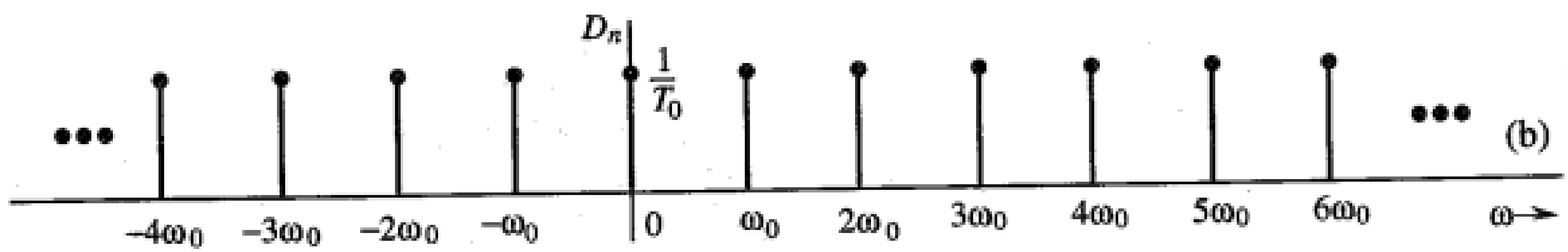


$$D_n = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} \delta(t) e^{-jn\omega_0 t}\, dt = \frac{1}{T_0}$$

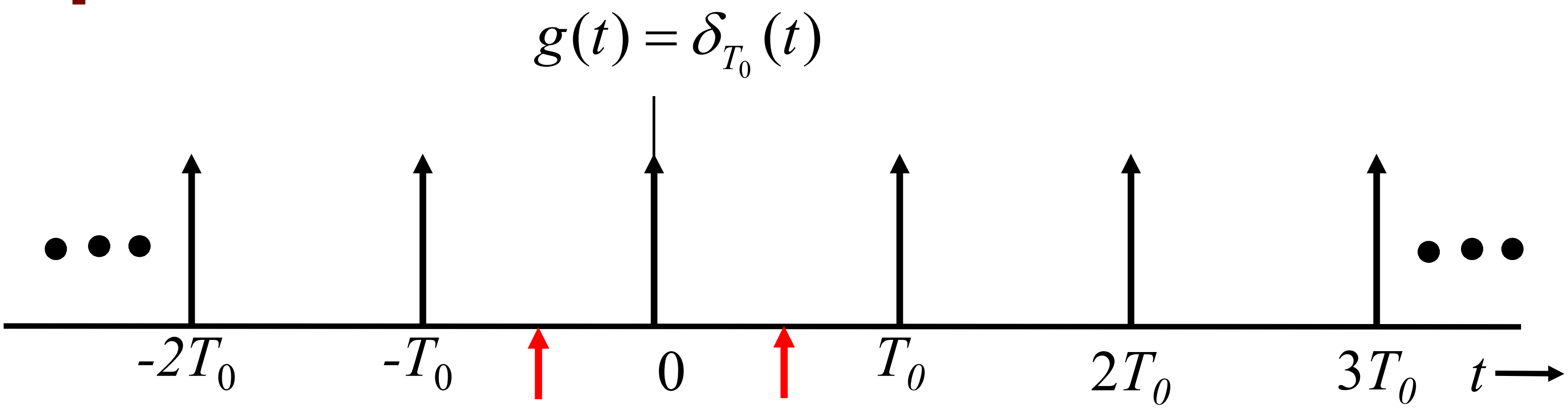
Fourier Series of Unit Impulse Train



$$D_n = \frac{1}{T_0}$$



Fourier Series of Unit Impulse Train



Fourier Series is

$$\delta_{T_0}(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t} = \frac{1}{T_0} \sum_{n=-\infty}^{\infty} e^{jn\omega_0 t}$$